NANYANG POLYTECHNIC
School of Engineering

Course	:	Diploma in Electronic & Computer Engineering (EGDF20)
Module	:	EGE322 – IoT System Project

Assignment 3 – MQTT Cloud Dashboard Exploration (Jigsaw Learning)

Introduction

In this assignment, you will explore MQTT communication using the ESP32 and various IoT cloud dashboard platforms.

During previous lab sessions:

- REST API communication was introduced using the ThingSpeak platform.
- MQTT communication concepts were introduced using Adafruit IO.

For this assignment, you will further strengthen your understanding of MQTT by exploring, debugging, documenting, and teaching the concepts to your peers.

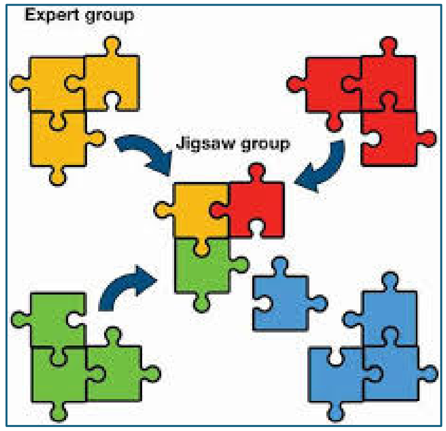
You will be provided with "**Instruction and Codes for MQTT communication using the ESP32 and the ThingSpeak platform"**. However, the guides provided may be *incomplete, unclear,* or *contain bugs.* Part of this assignment is for students to troubleshoot, debug, improve, and organize the materials into a clearer and more professional guide.

No additional external hardware or sensors are required for this assignment.
The ESP32 will internally generate sample data and upload the data to an IoT cloud dashboard using MQTT.

The purpose of this assignment is not only to learn one platform, but also to develop the ability to independently explore new technologies and platforms.

Jigsaw Learning Method

This assignment uses the **Jigsaw** Learning Method.



Students will work in pairs:

- Student A
- Student B

Each student will independently explore a different task or IoT dashboard platform. After completing the exploration:

- The student must create a clear tutorial or guide.
- The partner should be able to successfully complete the task by following the guide provided.

Afterward, both students will exchange knowledge and learn from each other.

This method encourages:

- Independent learning
- Communication skills
- Peer teaching
- Technical documentation skills
- Exploration of new technologies

Teaching others is one of the most effective ways to strengthen your own understanding.

"Instructions and Codes for MQTT communication with ESP32 and ThingSpeak"

1. Instruction

Step 1: Create a ThingSpeak account

Step 2: Create a new channel

Step 3: Enter a new channel name 'ESP32' and a new Field 'Count' in Field1.

Step 4: Click on 'Save Channel' to save the channel.

Step 5: Write down your Channel ID of the 'ESP32' channel

Step 6: Click on Devices and select MQTT

Step 7: Add a Device

Step 8: Download the credentials as a .txt file

Client ID: Cd4jKyiowSzw; Username: Xg4WKywg; Password: ARSQZVXeT+

Step 9: Create a new **micropython script** as follow main.py and then run with Thonny.

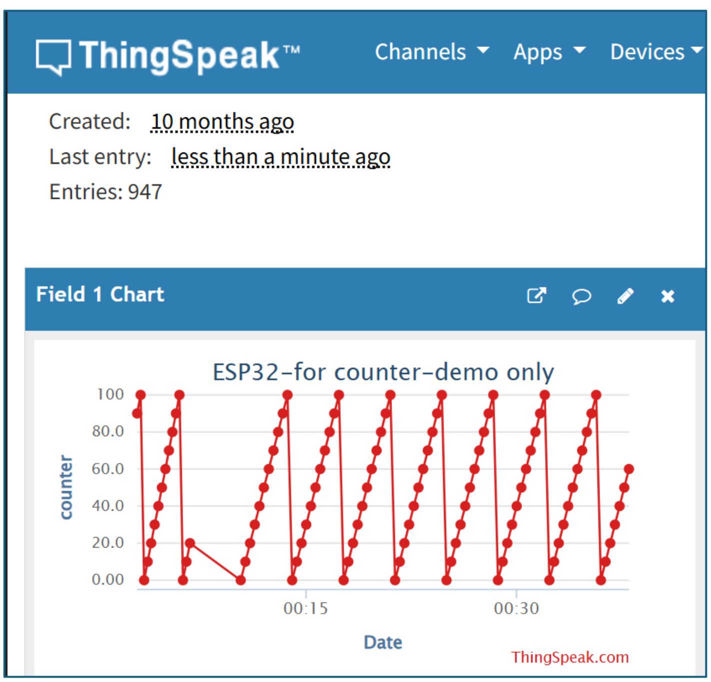
2. Codes

Main.py

```
import network
import time
from umqtt.simple import MQTTClient
from machine import Pin
WiFi_SSID = "xxxx"
WiFi_Password = "yyyy"
SERVER = "mqtt-v2.thingspeak.com"
PORT = 1883
CHANNE_ID = "Cd4jKyiowSzw#uHxIYNiE4JBM"
WRITE_API = "Xg4WKywgNzwukxJPYNiE4JBM"
CLIENT_ID = "DyAnDTMgDgsvLwMTHzwhCwg"
PASSWORD = "XAajr84Q5tBK2s15fnTYBM4T"
counter = 0
# Create topic to publish the message
topicOut = "channels/" + CHANNEL_ID + "/publish"
def wifi_connect():
    wlan = network.WLAN(network.STA_IF)
    wlan.active(True)
    if not wlan.isconnected():
        print("Connecting to network...")
        wlan.connect(WiFi_SSID, WiFi_PASS)
        while not wlan.isconnected():
```

```
            pass
    print("Connected to Wifi Router")
# Connect ESP32 to Wifi
wifi_connect()
# Create a client and connect to the MQTT broker
client = MQTTClient(CLIENT_ID, SERVER, PORT, USER, PASSWORD, keepalive=60)
client.connect()
print("MQTT connected")
while True:
    #client.check_msg()
    if counter > 100:
        counter = 0
    # Publish the topic message to the broker
    client.publish(topicOut, "field1=" + str(counter))
    print(counter)
    counter = counter + 10
    time.sleep(2)
```

3. IOT Dashboard



Assignment Tasks

Basic Task A – ThingSpeak MQTT Exploration

Student A Responsibilities

- Study and debug the provided MQTT sample codes for ThingSpeak.
- Understand how the ESP32 connects to the MQTT server.
- Understand how data is generated and uploaded to the dashboard.
- Improve and organize the original instructions into a clearer guide.
- Create a step-by-step tutorial for Student B.

The guide should include:

- Step-by-step procedures with screenshots
- Corrected **Micropython codes** with comments and troubleshooting notes
- Screenshots of the IoT dashboard results

Basic Task B – Adafruit IO MQTT Exploration

Student B Responsibilities

- Explore how to use MQTT with the **Adafruit IO platform.**
- Repeat a similar MQTT data upload task using ESP32.
- Create a clear tutorial for Student A to follow.

The guide should include:

- Step-by-step procedures with screenshots
- Corrected **Micropython codes** with comments and troubleshooting notes
- Screenshots of the IoT dashboard results

Exploration Tasks

Groups are encouraged to explore additional IoT dashboard platforms to broaden their learning experience. Additional exploration and successful implementation may contribute to higher marks.

Self-Exploration IoT Dashboard Platforms (Choose Any One):

- ThingsBoard
- Tago IO
- Home Assistant
- Node-Red Dashboard
- Google Firebase
- Ubidots
- Blynk
- AllThingsTalk Maker

Learning Resources

Students are encouraged to explore and learn independently using various online resources, including:

- AI tools
- YouTube tutorials
- Official documentation websites
- GitHub repositories
- Professional maker/tutorial websites such as:
  - Hackster.io
  - Cytron Technologies
  - DFRobot
  - Random Nerd Tutorials

In the current AI era, learning and exploration have become more accessible than ever. The ability to independently learn new technologies is an important future skill.

Peer Assessment

After completing the assignment:

- Student A and Student B will evaluate each other by online form.
- Assessment will follow the rubrics provided by the lecturer.
- Peer evaluations will remain confidential.
- The lecturer will also review the submitted work and provide additional grading.

Exercise Submission

1) Submit the guide created during the Jigsaw learning activity in Microsoft Word format.
    a. Step-by-step procedures with screenshots
    b. Corrected **Micropython codes** with comments and troubleshooting notes
    c. Screenshots of the IoT dashboard results
    d. Any limitations encountered during the exploration (if applicable)

2) Submit your completed “Basic Task Guide” or “Exploration Task Guide” to Brightspace.

3) Do not forget to complete and submit the online peer evaluation form.